NoveltyBench: Evaluating Language Models for Human-like Diversity

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Abstract

Language models have demonstrated remarkable capabilities on standard benchmarks, yet they struggle increasingly from *mode collapse*, the inability to generate diverse and novel outputs. Our work introduces NoveltyBench, a benchmark specifically designed to evaluate the ability of language models to produce multiple distinct and high-quality outputs. NoveltyBench utilizes prompts curated to elicit diverse answers and filtered real-world user queries. Evaluating 20 leading language models, we find that current state-of-the-art systems generate significantly less diversity than human writers. Notably, larger models within a family often exhibit less diversity than their smaller counterparts, challenging the notion that capability on standard benchmarks translates directly to generative utility. While prompting strategies like in-context regeneration can elicit diversity, our findings highlight a fundamental lack of distributional diversity in current models, reducing their utility for users seeking varied responses and suggesting the need for new training and evaluation paradigms that prioritize diversity alongside quality.[1]

1 Introduction

Diversity of opinion, preference, and experience is a fundamental trait of being human. If you were to ask "Tell me a joke" or "What is the best book of all time?" to the next five people you talk to, with high likelihood you would receive five different answers. It is reasonable to expect language models to generate responses that have the same level of diversity as humans. Yet, when we ask models such as GPT-4 to recommend a movie or Claude 3 to suggest several vacation destinations, we often receive variations of the same few ideas—a phenomenon known as *mode collapse* Hamilton (2024).

This lack of diversity in language model outputs represents a significant limitation. Today's aligned language models tend to produce lower entropy distributions than earlier generations of models (Zhang et al., 2024b), and when asked to generate (using random sampling) several responses to an open-ended prompt, the responses will often contain substantial near-duplicates (O'Mahony et al., 2024). This tendency can harm the utility of these models for subjective tasks where different users may have diverging preferences and needs (Zhang et al., 2024a). Sorensen et al. (2024) refer to the inability of today's LLMs to produce diverse generations as a failure of *pluralistic alignment*, which can lead to less useful and customizable AI systems.

While today's LLMs are evaluated at great length for knowledge and reasoning abilities, they are rarely evaluated for response diversity, and there are no widely-adopted benchmarks for assessing this trait. The majority of existing evaluation benchmarks are "mode-seeking", in the sense that they only assess the quality of the most likely generation of language

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[1]Code and data are available at https://novelty-bench.github.io.

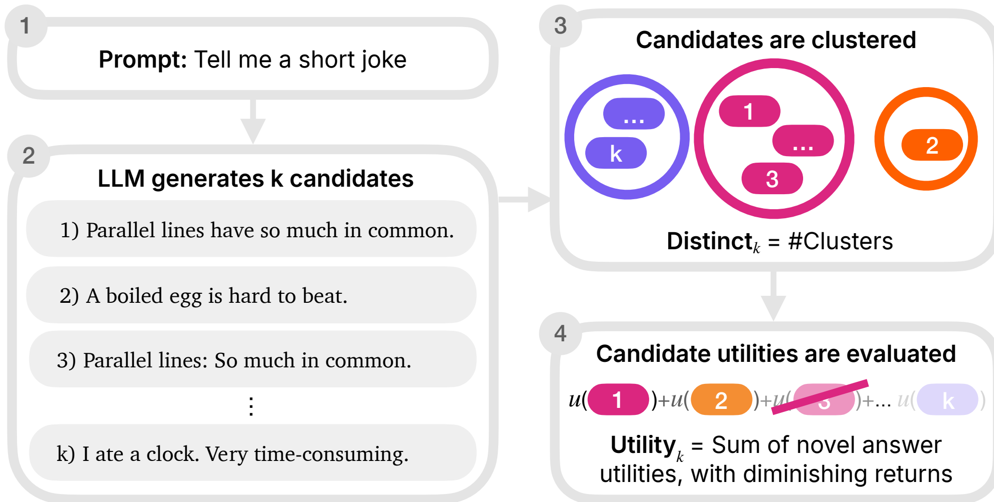

Figure 1: NOVELTYBENCH overview. We partition generations into functional equivalence classes and introduce two metrics: distinct$_k$ counts unique equivalence classes among k samples, and utility$_k$ combines novelty and quality, weighing utility of individual generations by user patience ($p = 0.8$) and only considering *novel* generations.

models and do not assess the capability of models to produce meaningful alternatives.[2] This evaluation paradigm is problematic because it creates misaligned incentives: model developers focus on improving the quality of the single most likely generation rather than the diversity of the entire distribution of possible outputs.

In this work, we set out to measure not only what language models can generate, but also what they *cannot*. To this end, we propose NOVELTYBENCH (Figure 1), a benchmark for measuring how well language models can generate multiple differing but still correct answers to user requests that involve subjectivity, randomness, and creativity—in other words, queries that a room of humans would produce a large variety of answers to. We intend for our benchmark to serve as a complement to existing quality-based evaluation benchmarks, encouraging model developers to strive toward more pluralistic models, while also taking generation quality into account.

In summary, we introduce NOVELTYBENCH, a benchmark consisting of 1,100 prompts that we expect to elicit variable answers from humans, 100 of which are annotated with responses from eight human annotators. We propose a unified measure of novelty and quality which can be used to gauge how well an LLM can produce diverse, high-quality responses to a given prompt. We evaluate NOVELTYBENCH on twenty frontier models, including GPT-4o, Claude 3.5 Sonnet and Gemini 2.0 Pro, and find that all suffer from a lack of diversity. Our evaluation further reveals that larger and more capable models in the same model family tend to produce less diverse outputs, highlighting a concerning trend in the development of state-of-the-art language models. These findings underscore the need for future work on training techniques that promote diversity alongside quality, and for evaluation paradigms that better reflect the full spectrum of human preferences and creative capabilities.

2 Related Work

Prior work has established that language model outputs often exhibit biases towards specific demographic groups (Santurkar et al., 2023; Sun et al., 2023), genders (del Arco et al., 2024), countries (Durmus et al., 2023), or personas (Giorgi et al., 2024; Oketunji et al., 2023). This homogenization can limit LLM utility on downstream tasks (Agarwal et al., 2024;

[2]In an analysis of 67 benchmark papers published at COLM 2024 and ICLR 2025, we found that over 90% of these papers evaluate language models based on a single/best generation, instead of considering the full model output distribution.

Song et al., 2024). Research into diversifying language model outputs has seen several promising directions. Common approaches include adjusting decoding methods, such as increasing temperature (Shur-Ofry et al., 2024; Peeperkorn et al., 2024), or modifying training objectives (Li et al., 2016; Zhang et al., 2024b; Li et al., 2024; Lanchantin et al., 2025).

The idea that we should evaluate language models not only on the quality of their generations but also on their diversity is not entirely new. The 2020 Duolingo Shared Task involved generating several diverse but still correct translations of an input sentence (Mayhew et al., 2020). Recent work by Cheng et al. (2024) further highlights the probabilistic nature of commonsense knowledge in language models, and proposed an probabilistic evaluation method for commonsense knowledge that relies on open-ended generation instead of multiple-choice questions. Numerous studies of NLG systems have proposed metrics for evaluating diversity, based on perplexity (Lewis et al., 2017; Fan et al., 2018), n-gram overlaps (Zhu et al., 2018; Shu et al., 2019; Dušek et al., 2020), and embedding distances (Du & Black, 2019). However, existing metrics often correlate poorly with human judgments of content diversity (Tevet & Berant, 2021). In contrast to prior work, NOVELTYBENCH uniquely evaluates an LLM's capacity to generate a set of outputs that are simultaneously diverse and high-quality.

3 Benchmark

Our benchmark evaluates language model diversity using two distinct datasets: NB-CURATED which contains 100 prompts manually curated by the authors, and NB-WILDCHAT which consists of 1,000 prompts automatically curated from real user interactions with ChatGPT. LLM performance on the benchmark is evaluated based on the quality and novelty of the set of generations produced by the LLM.

3.1 Dataset Curation

To test language models' capabilities to produce novel generations, we need a dataset of prompts that elicit diverse responses. We developed two sets of such prompts: NB-CURATED contains prompts designed by the authors to have multiple valid answers, while NB-WILDCHAT is a collection of 1,000 prompts sourced from the WildChat-1M dataset (Zhao et al., 2023).

NB-CURATED. The curated dataset is designed to contain four distinct categories where diversity is expected.

- **Randomness**: prompts that involve randomizing over a set of options.
  Example: `Roll a make-believe 20-sided die.`
- **Factual Knowledge**: prompts that request underspecified factual information, which allow many valid answers.
  Example: `List a capital city in Africa.`
- **Creative Writing**: prompts that involve generating a creative form of text, including poetry, and story-writing.
  Example: `Tell me a riddle.`
- **Subjectivity**: prompts that request subjective answers or opinions.
  Example: `What's the best car to get in 2023?`

The authors of the paper then write responses to each of the prompts, resulting in 8 human responses per prompt. Since this is a rather homogeneous set of human annotators (all work/study at the same institution), we expect this response set to represent an approximate *lower* bound on actual human diversity. However, as we will show in Section 4.3, even this group produced more diverse responses than most state-of-the-art language models.

NB-WILDCHAT. WildChat is a collection of 1M real user conversations with ChatGPT, which is a valuable resource for understanding *real user needs of creativity* since they represent usage of language models in the wild. However, we cannot use this dataset directly because